

Do Circuits Survive Compression? A Null-Model Audit of Circuit Stability Under Quantization and Pruning

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Abstract

Does compression alter computation beyond the variability of circuit extraction? We develop a null-first pipeline using a Circuit Stability Index (CSI), exact-edge and routing-level comparisons, and a frozen-null preregistration protocol. Synthetic validation passes 459 tests (one skipped), yielding exact-edge CSI 0.333 with interval $[0.25, 0.40]$. Reproduced feature baselines favor Wanda over magnitude pruning; fixed-dictionary Jaccard correlates with output damage at $\rho = -0.975$. Real-model CSI and causal confirmation remain pending; the null is not yet frozen.

1 Introduction

Feature-level compression studies motivate a complementary audit of computation [1, 2]. Circuit size can vary by approximately $40\times$ with extraction thresholds, and apparent stability depends on whether comparisons use exact edges or routing structure [3, 4]. Activation patching also mixes direct and interaction effects, $NIE = PIE + INT$ [5]. We target Gemma-2-2B, Llama-3.2-1B, and Pythia on indirect object identification (IOI), greater-than, and docstring tasks.

2 Methodology

For B extraction configurations and S seeds, let f denote the vector of edge-selection frequencies $s(e)$. For compression condition c and task T , we measure

$$s(e) = \frac{\#\{\text{graphs containing } e\}}{BS}, \quad D(c) = \text{Dist}(f_{\text{dense}}, f_c), \quad \text{CSI}(c, T) = \frac{D(c)}{\text{median}(D_{\text{null}}(c))}.$$

- 1. Null-first execution.** Stage A builds the dense reference; Stage B generates R nulls and irreversibly freezes their hash. Stage C computes distances and CSI only after this guard passes; Stage D performs causal patching and cross-audit.
- 2. Matched controls and ensemble.** Magnitude-matched random perturbations preserve the per-tensor Frobenius norm of compression-induced ΔW ; an alternative perturbation family matches perplexity within 0.05. The planned pilot uses $B = 16$, $S = 5$, $R = 20$ on a non-nested threshold grid; synthetic validation uses (4, 2, 3).
- 3. Extraction and comparison.** Attribution-patching and edge-pruning graphs are complemented by a dictionary-free dense-node variant (node/edge thresholds: 0.8/0.98). We compare exact edges and routing heads, with a strict guard against empty routing projections. Primary distance is L1 over the edge union, with Jensen–Shannon and normalized-L1 ablations.
- 4. Uncertainty and interpretation.** The protocol specifies 1,000 bootstrap replicates over configurations, seeds, and null draws. $\text{CSI} \gg 1$ with a confidence interval excluding 1 supports selective damage; $\text{CSI} \approx 1$ is consistent with null variability; $\text{CSI} < 1$ indicates less change than the null. Audits include Spearman cross-comparison, a 0.0–0.9 threshold sweep, chance floors, interaction checks, and Benjamini–Hochberg correction at $q = 0.05$.

3 Results

Reproduced baselines. Dense perplexities are 8.21 (Gemma-2-2B), 9.99 (Gemma-3-1B), and 9.28 (Llama-3.2-1B). At 30% sparsity on Gemma-2-2B, Wanda retains MNN survival of **0.794 versus 0.210** for magnitude pruning, and Jaccard of 0.591 versus 0.317 (Table 1). Downstream accuracy falls from 0.759 (dense) to 0.325 (magnitude, 50%) and 0.627 (Wanda, 50%). The seed MNN floor is 0.0273; Kendall’s $W = 0.9416$.

Comparison-level disagreement. Reproduced exact-edge overlaps are 0.163/0.142 against chance 0.126/0.117 ($p = 0.040/0.106$); routing overlaps are 0.666/0.554 against 0.199/0.185 ($p < 0.001$). Cross-seed Jaccard is 0.363. These results motivate reporting both levels rather than a single overlap score.

Synthetic validation and calibration. Exact-edge CSI is 0.333 [0.25, 0.40], with $D = 0.25$ and null median 0.75; routing CSI is 0.0 [0, 0], with $D = 0$. Exact-edge Jaccard is 0.714 versus a chance floor of 0.0 [0, 0.091]; routing Jaccard is 1.0 versus 0.0 (both $p = 0.002$). Separately, feature calibration identifies fixed-dictionary Jaccard as the strongest correlate of output damage among the reported diagnostics (Table 2).

Table 1: Pruning baselines: sparsity s , perplexity (PPL), MNN survival, and fixed-dictionary Jaccard. Dashes indicate unreported values.

Model	Method	s	PPL	MNN	Jacc.
Gemma-2-2B	Magn.	0.3	13.08	0.210	0.317
	Magn.	0.5	377.03	0.002	0.140
	Wanda	0.3	8.82	0.794	0.591
	Wanda	0.5	13.47	0.148	0.314
Llama-3.2-1B	Magn.	0.5	2412.44	0.511	—
	Wanda	0.3	10.65	0.850	—
	Wanda	0.5	33.35	0.709	—

Table 2: Feature calibration: Spearman correlation with log-perplexity change, 95% bootstrap interval, and sample size.

Diagnostic	ρ	95% CI	n
Fixed-dict Jacc.	−0.975	[−1.00, −0.873]	17
Downstream acc.	−0.906	[−0.991, −0.632]	16
MNN	−0.684	[−0.888, −0.276]	17
FVU	0.556	[0.030, 0.918]	17
Fragility	−0.032	[−0.473, 0.514]	17

Contribution and current limits. The implementation combines bootstrapped CSI, matched-null harnesses, frequency bands, routing comparisons, chance floors, interaction flags, and freeze-as-preregistration. The project maps 50 science, 33 experiment, and 28 compression nodes across 22 communities. These are infrastructure and reproduction results: **no real-model CSI has been measured**, and causal confirmation and cross-audit remain pending.

4 Future Scope

- Pin data and prompts (4 h).** Record Hugging Face revisions, 200–500 prompts per task, and the WikiText-2 protocol (window 1,024; stride 512) to protect the dense reference. Deliver confirmed task configurations and a passing protocol test.
- Finalize preregistration (0.5 h).** Lock L1 and Jensen–Shannon choices and resample over B, S, r rather than seeds alone. Require all 14 distance tests to pass before freezing the null.
- Set consensus bands (0.5 h).** Use core $s = 1$ and noise $s = 7/16$ at $B = 16$ on the crossed grid, following CIRCUS; check Jaccard ≈ 0.33 and pass decomposition/grid tests.
- Run compression experiments (2 weeks).** Evaluate RTN INT8/INT4, GPTQ [7], AWQ, magnitude pruning, and Wanda [6] against frozen nulls. Deliver CSI tables with confidence intervals at both comparison levels.
- Complete causal audit and paper (2 weeks).** Use grouped patching with an interaction split, cross-audit, and threshold sweeps to link circuit changes to behavior. Deliver an AAI-UC draft and a one-command rebuild.

References

- [1] A. Borobia et al., *How Pruning Reshapes Features*, arXiv:2603.25325, 2026.
- [2] E. Duan, *Perplexity Can Miss SAE Feature Damage Under Quantization*, arXiv:2606.03002, 2026.
- [3] S. Parekh, *CIRCUS: Circuit Consensus under Uncertainty via Stability Ensembles*, arXiv:2603.00523, 2026.
- [4] S. Sheng and F. Fu, *Circuit Claims Depend on What Is Extracted and How Compared*, arXiv:2607.18921, 2026.
- [5] V. Vaidyanathan et al., *Curse of Multiple Mediators: Hidden Interaction Effects in Activation Patching*, arXiv:2606.27510, 2026.
- [6] M. Sun et al., *Wanda: Simple Effective Pruning*, arXiv:2306.11695, 2024.
- [7] E. Frantar et al., *GPTQ*, arXiv:2210.17323, 2023.